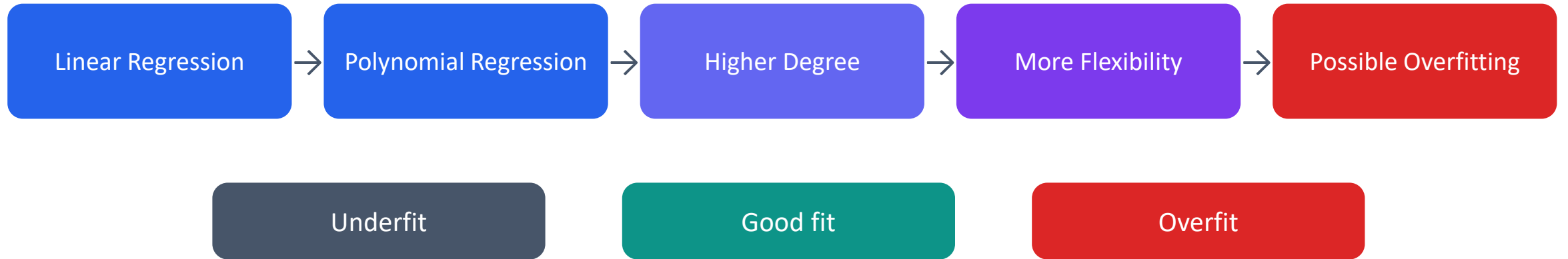


Regularization Techniques in Machine Learning

Controlling Model Complexity with Ridge, Lasso & Elastic Net



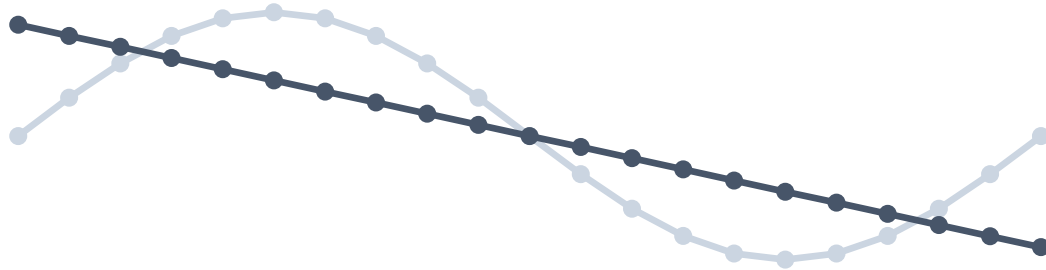
Let's Start Where We Left Off



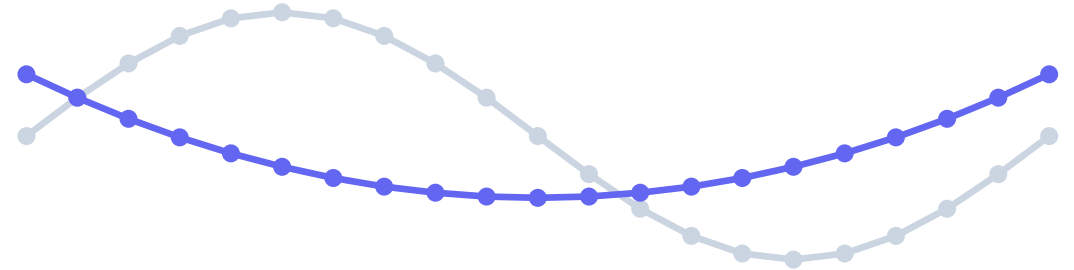
“If a model becomes too complex, how can we control it?”

The Problem: Model Complexity

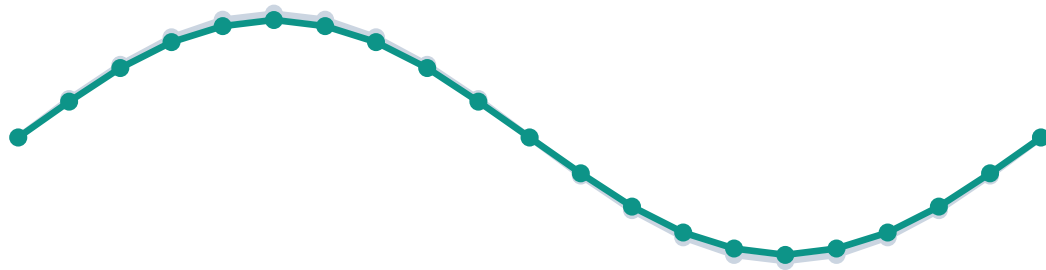
Degree 1 · high bias



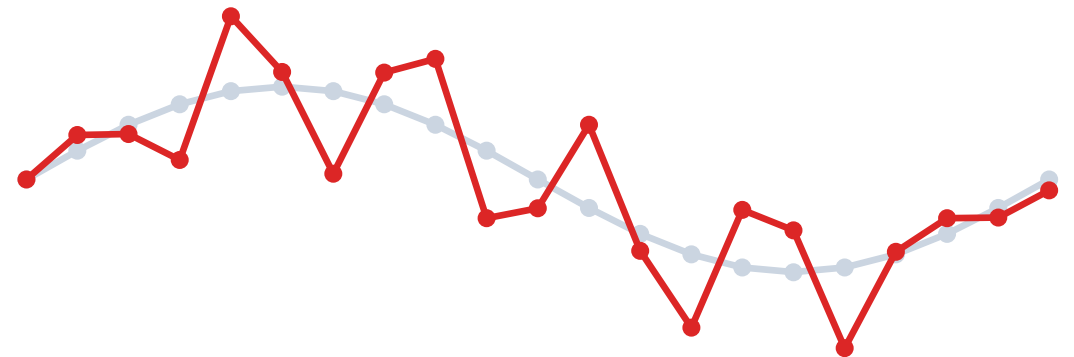
Degree 2



Degree 5 · good fit

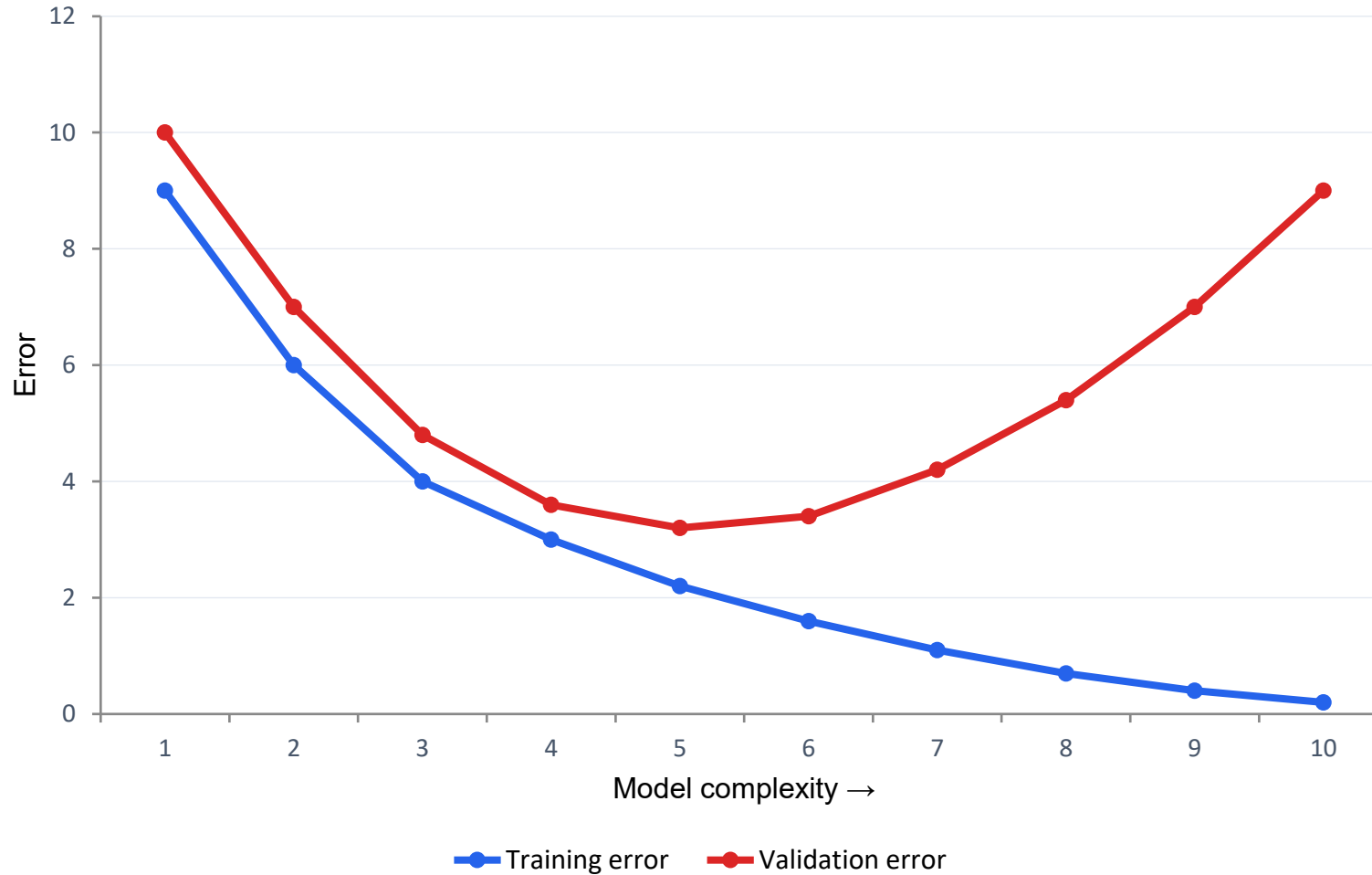


Degree 15 · high variance



Illustrative sketch — curves drawn for teaching, not fitted to data

Training Error vs Generalization



Minimizing training error alone is not enough.

Sweet spot = lowest validation error.
Regularization helps us stay near it.

Illustrative teaching example — synthetic values, not real data

The Big Idea: Regularization

Can we fit the data well WITHOUT letting the model become unnecessarily complex?

$$\text{Original objective (RSS)} + \text{Complexity penalty} = \text{Regularized objective}$$

Large coefficients = unstable, wiggly models. Discourage them, and you trade a little bias for lower variance. The intercept is typically not penalized.

Ridge Regression — L2 Regularization

$$J(\beta) = RSS + \lambda \sum_j \beta_j^2$$

$\hat{y} = X\beta + \beta_0$ · $\lambda \geq 0$ = strength · β_j = coefficients · β_0 not penalized

$\lambda = 0$

Ordinary least squares

λ small

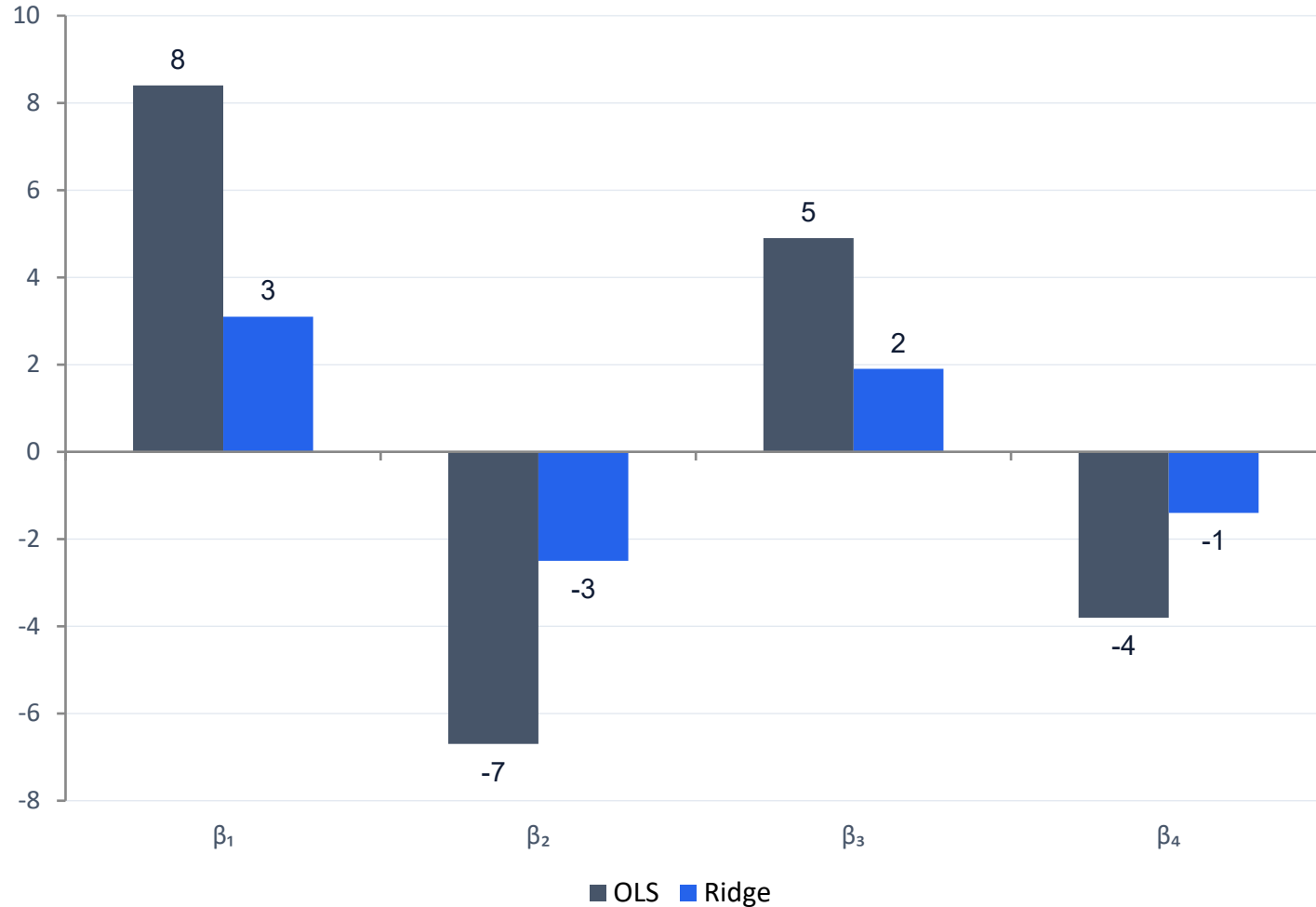
Mild shrinkage

λ large

Strong shrinkage; risk of underfitting

Scikit-learn: Ridge(alpha= λ). Ridge uses RSS without a $1/n$ factor; Lasso and ElasticNet use $(1/2n) \cdot \text{RSS}$. Convention differences change the scale of alpha, not the idea.

Ridge Shrinks Coefficients Toward Zero



Ridge shrinks coefficients toward zero.

Usually NOT exactly zero.

Large coefficients $\rightarrow$ sensitive to small input changes $\rightarrow$ high variance.
Shrink them $\rightarrow$ lower variance, slightly higher bias.

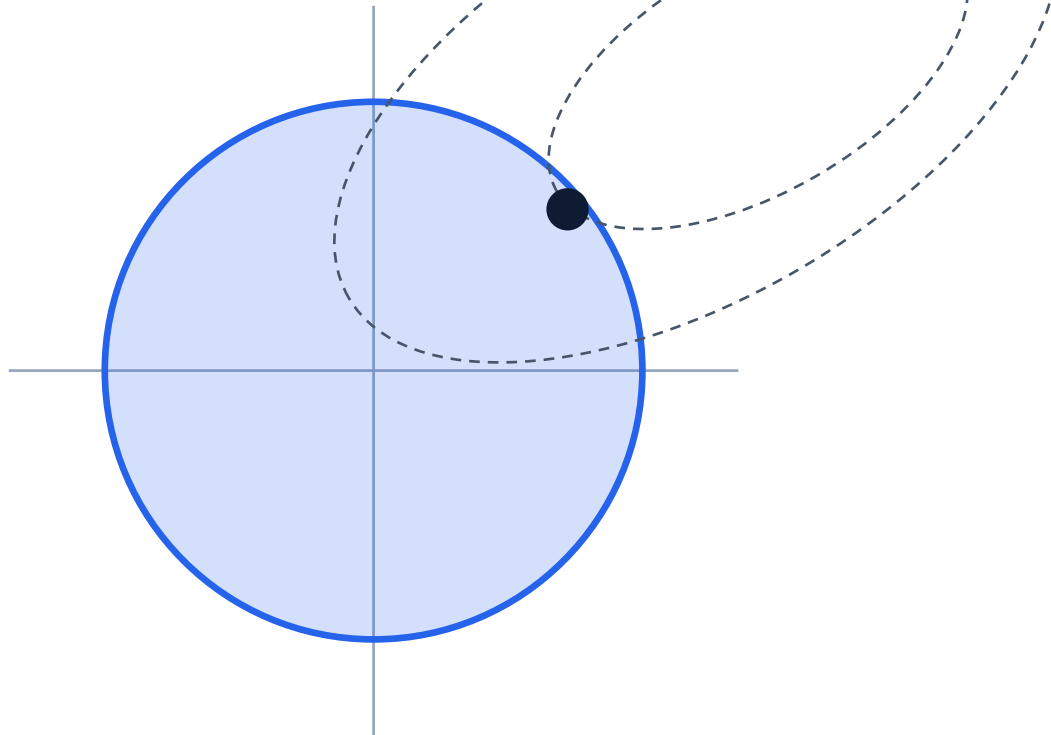
OLS — Ordinary Least Squares

OLS is the standard method used to find the best-fitting coefficients in Linear Regression.

Geometry: Why L1 Gives Zeros, L2 Does Not

$$\text{L2: } \beta_1^2 + \beta_2^2 \leq t$$

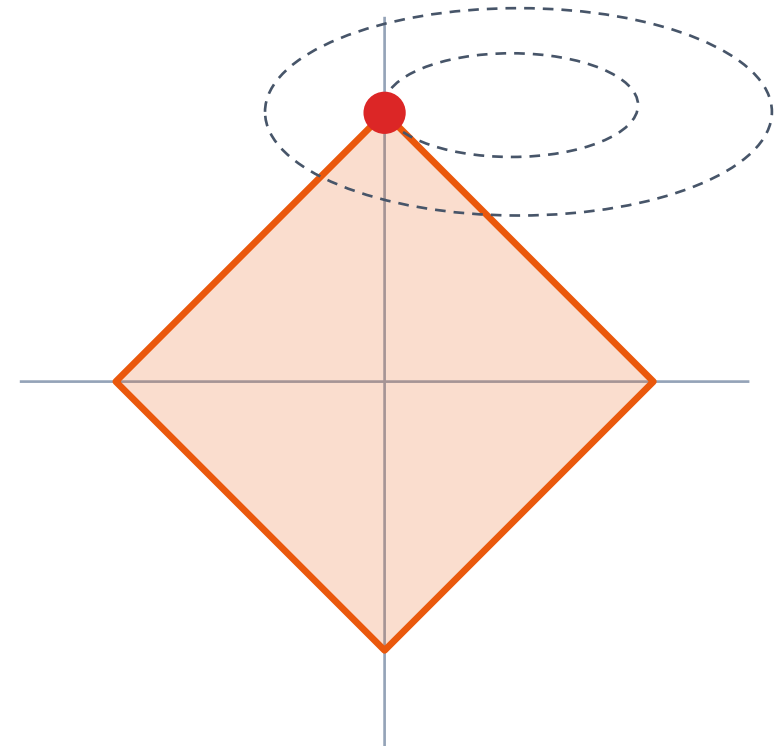
Ridge · the L2 region is a circle



Smooth boundary: contact rarely on an axis

$$\text{L1: } |\beta_1| + |\beta_2| \leq t$$

Lasso · the L1 region is a diamond



Corner on an axis: $\beta_1 = 0$ becomes likely

Dashed ellipses = loss contours (schematic)

Lasso Regression — L1 Regularization

$$J(\beta) = RSS + \lambda \sum_j |\beta_j|$$

OLS

4.2

-3.7

2.1

0.8

-0.4

Lasso

2.6

0

1.1

0

0

Lasso can perform implicit feature selection.

Illustrative coefficient vectors

Ridge vs Lasso: The Key Difference

RIDGE — L2

Penalty

$$\sum \beta_j^2$$

Effect

Shrink coefficients

Exact zeros?

Usually no

Feature selection

No explicit sparsity

LASSO — L1

Penalty

$$\sum |\beta_j|$$

Effect

Shrink coefficients

Exact zeros?

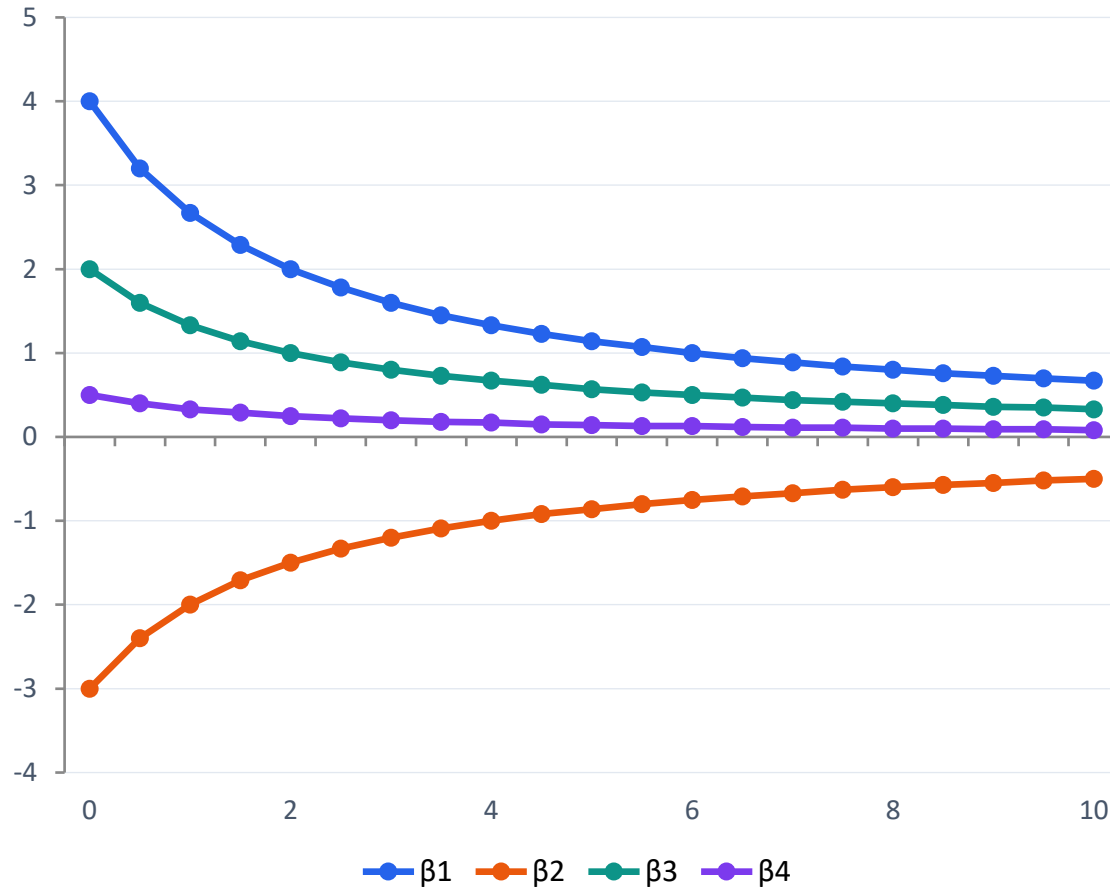
Yes

Feature selection

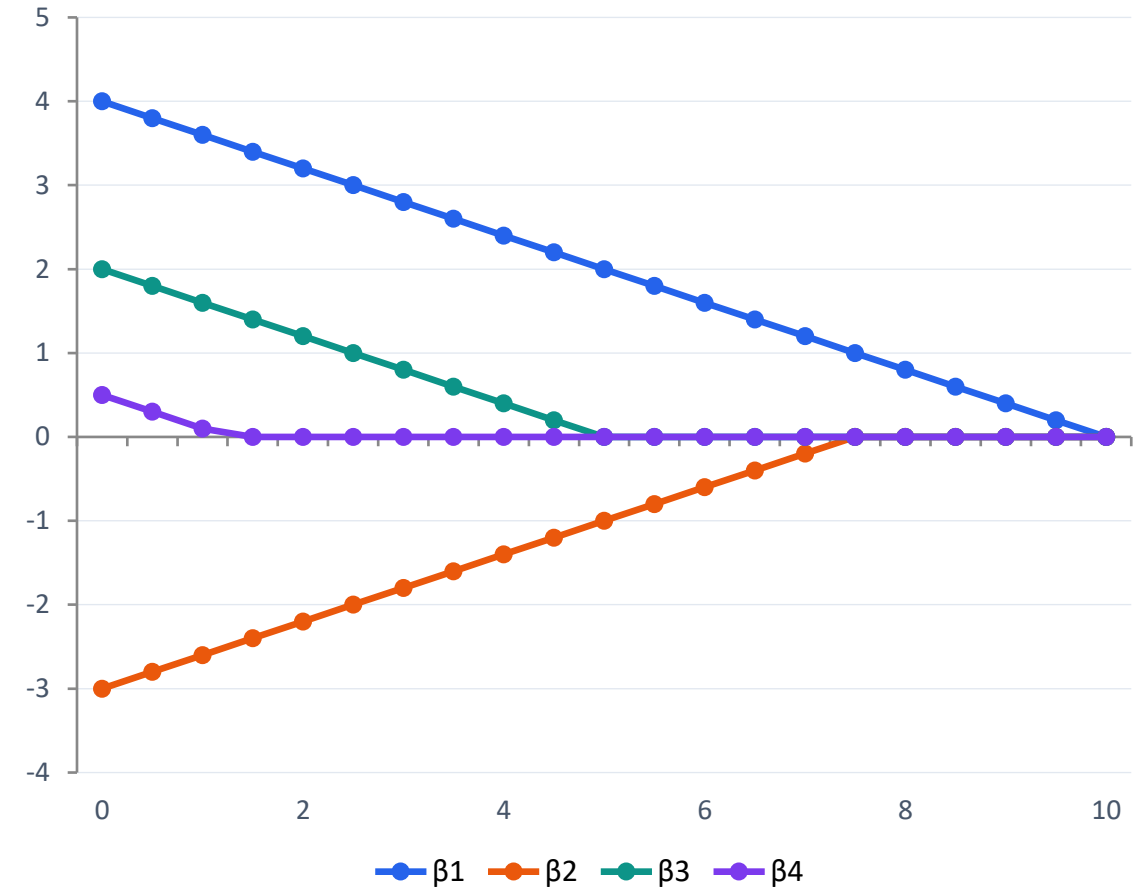
Yes

Coefficient Paths: Ridge vs Lasso

Ridge: smooth shrinkage

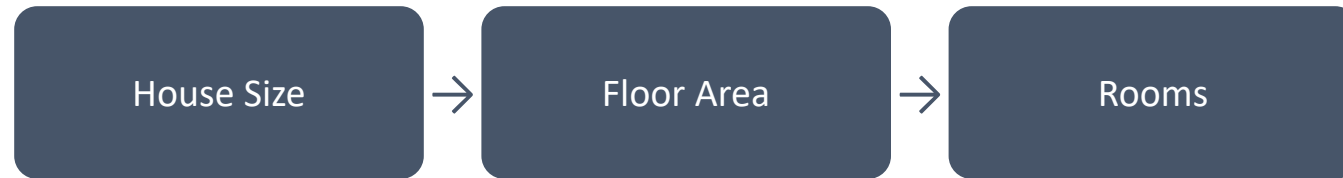


Lasso: coefficients hit zero



Illustrative paths (x-axis: regularization strength λ). Synthetic formulas for teaching.

Correlated Features



highly correlated

Ridge

Tends to share weight across correlated features

Lasso

May keep one and zero the rest; the choice can be unstable

Practical caution: a feature Lasso drops is not necessarily unimportant.

Elastic Net — L1 + L2

$$J(\beta) = RSS + \lambda [\alpha \sum |\beta_j| + (1-\alpha) \sum \beta_j^2]$$

$\alpha = 0 \rightarrow$ Ridge



$0 < \alpha < 1 \rightarrow$ Elastic Net



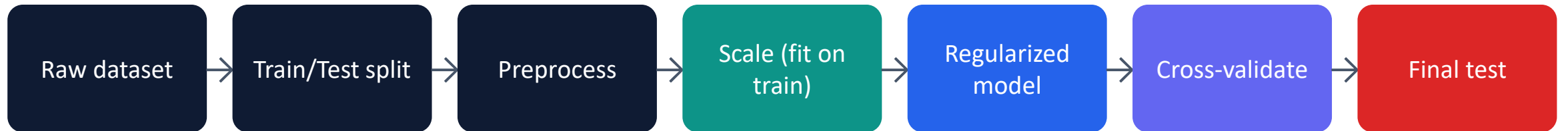
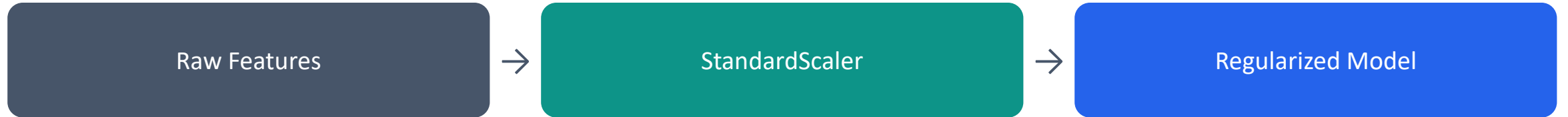
$\alpha = 1 \rightarrow$ Lasso

Our notation	scikit-learn
λ (strength)	alpha
α (L1 mix)	l1_ratio
$\alpha = 1 / \alpha = 0$	l1_ratio=1 (Lasso-like) / 0 (Ridge-like)

Ridge vs Lasso vs Elastic Net

	Ridge	Lasso	Elastic Net
Penalty	L2	L1	L1 + L2
Shrinkage	Yes	Yes	Yes
Exact zeros	Usually no	Yes	Yes
Feature selection	No	Yes	Yes
Correlated features	Shares weight	Can be unstable	Often better
Typical use	Stable prediction	Sparse model	Sparse + correlated
Main limitation	Keeps all features	Arbitrary pick	Two hyperparameters

Scale First, and Avoid Leakage



Scale features before Ridge, Lasso or Elastic Net.

Fit the scaler on training data only. Use a Pipeline.

Python: Ridge and Lasso

```
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Ridge

model = make_pipeline(
    StandardScaler(),
    Ridge(alpha=1.0)
)
model.fit(X_train, y_train)
```

```
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Lasso

model = make_pipeline(
    StandardScaler(),
    Lasso(alpha=0.1, max_iter=10000)
)
model.fit(X_train, y_train)
```

alpha = penalty strength. max_iter: Lasso may need more iterations to converge.

Python: Elastic Net and Tuning

```
pipe = make_pipeline(  
    StandardScaler(),  
    ElasticNet(max_iter=10000)  
)  
param_grid = {  
    "elasticnet__alpha": [0.001, 0.01, 0.1, 1, 10],  
    "elasticnet__l1_ratio": [0.1, 0.3, 0.5, 0.7, 0.9],  
}
```

```
search = GridSearchCV(  
    pipe, param_grid, cv=5,  
    scoring="neg_root_mean_squared_error",  
)  
search.fit(X_train, y_train)  
print(search.best_params_)
```

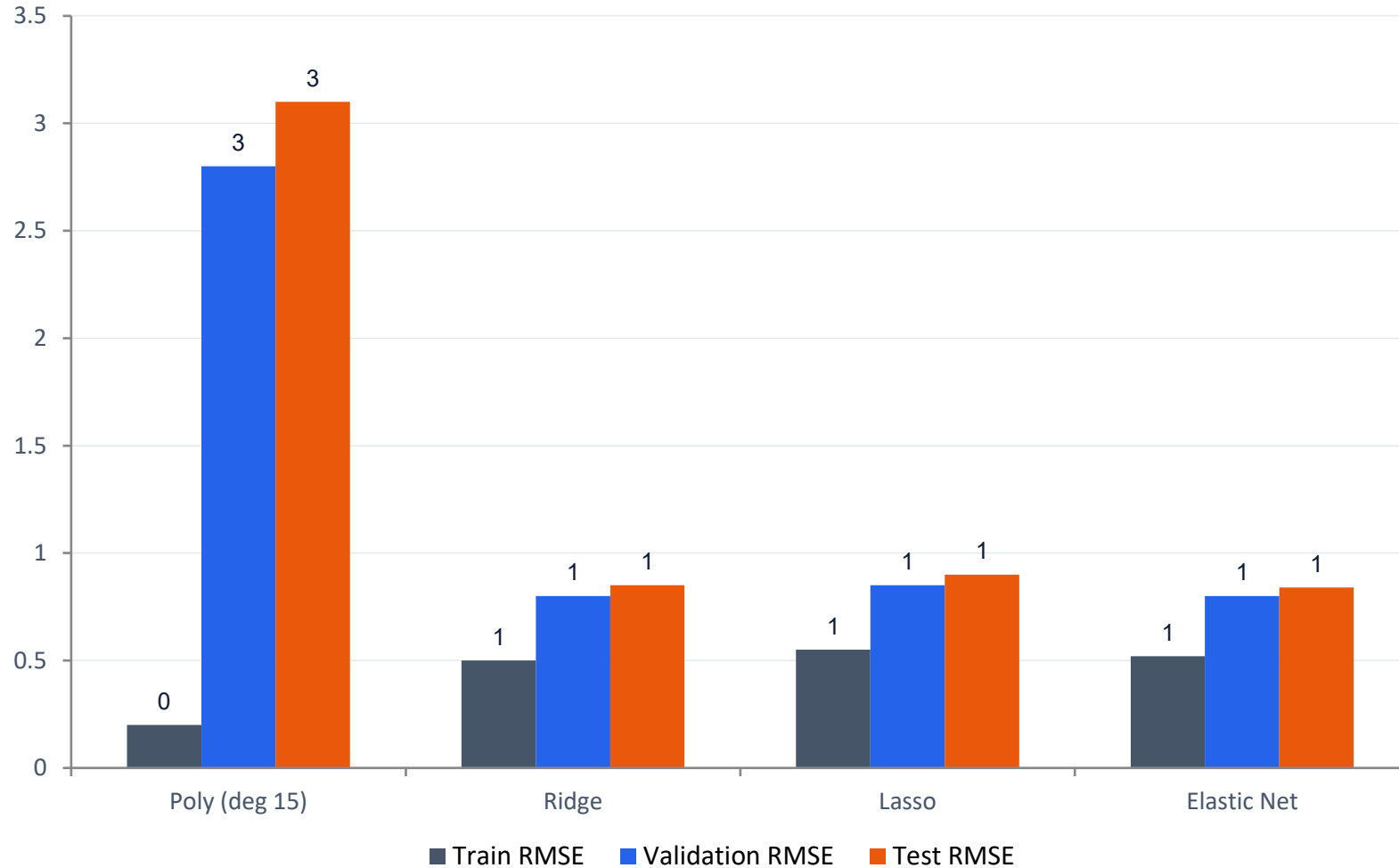
alpha → overall strength

l1_ratio → L1/L2 mixture

Log-spaced grid: strength matters on a multiplicative scale

Also: RandomizedSearchCV for big grids

Visual Experiment: Polynomial Features + Regularization



Plain polynomial: lowest train error, worst test error.

Regularization trades a bit of train fit for generalization.

Illustrative example — values generated for teaching.

Regularization Is Not “Always Better”

MYTH Ridge does feature selection



REALITY Ridge shrinks but usually does not give exact zeros.

MYTH Lasso always picks the right features



REALITY Selection can be unstable, esp. with correlated predictors.

MYTH More regularization is always better



REALITY Too much causes underfitting.

MYTH Scaling does not matter



REALITY Scaling is generally important for L1/L2.

When Should I Use What?

Coefficient shrinkage / many correlated features



Ridge

Sparse model / feature selection



Lasso

Sparsity + correlated predictors



Elastic Net

Interpretable and sparse



Lasso / Elastic Net

Practical heuristics, not absolute rules.

Classroom Challenge

You have 100 features.
30 are highly correlated.
You want a relatively sparse model.
What would you try first?

**A. Ordinary Least
Squares**

B. Ridge

C. Lasso

D. Elastic Net

Answer: Elastic Net, and Then Validate It

L1 → sparsity

L2 → stabilizes correlated
predictors

Combination → useful
compromise

The correct choice should ultimately be validated empirically using cross-validation.

Practical Checklist



Split data properly



Prevent data leakage



Scale features



Choose regularization family



Tune hyperparameters using
CV



Evaluate on held-out data



Inspect coefficients



Check stability



Compare against a baseline

One-Slide Cheat Sheet

Technique	Penalty	Exact zeros?	Feature selection	Correlated predictors
Ridge	L2	Usually no	No	Strong
Lasso	L1	Yes	Yes	Can be unstable
Elastic Net	L1 + L2	Yes	Yes	Often useful

Ridge → shrink

Lasso → shrink + select

Elastic Net → shrink + select + stabilize

Don't just fit the data. Control the complexity of the model.

Ridge • Lasso • Elastic Net

Regularization controls complexity • L2 shrinks • L1 can zero out coefficients • Elastic Net combines both • scale features • tune with cross-validation